Design Fair Presentation

Team 2025148

400G Ethernet Exploration — An FPGA-Accelerated Cloud Platform for BER Analysis

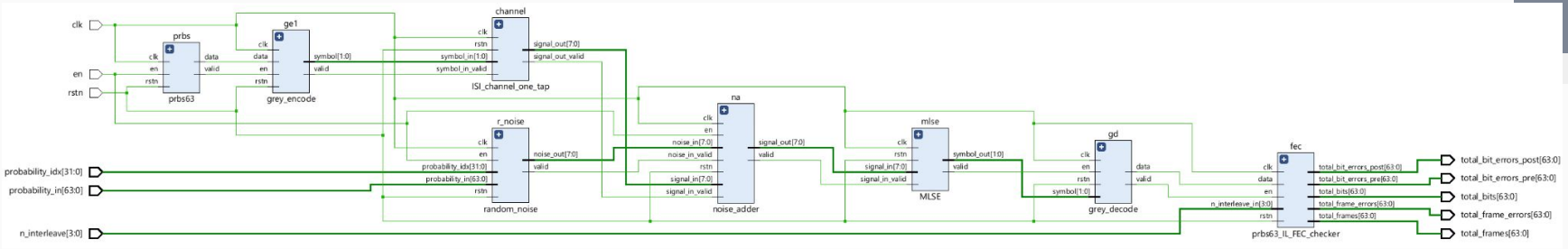
Elizabeth Tyra Sumual
Jiaxin (Brett) Yang
Kevin Jiao
Suet Ling (Celine) Kwan

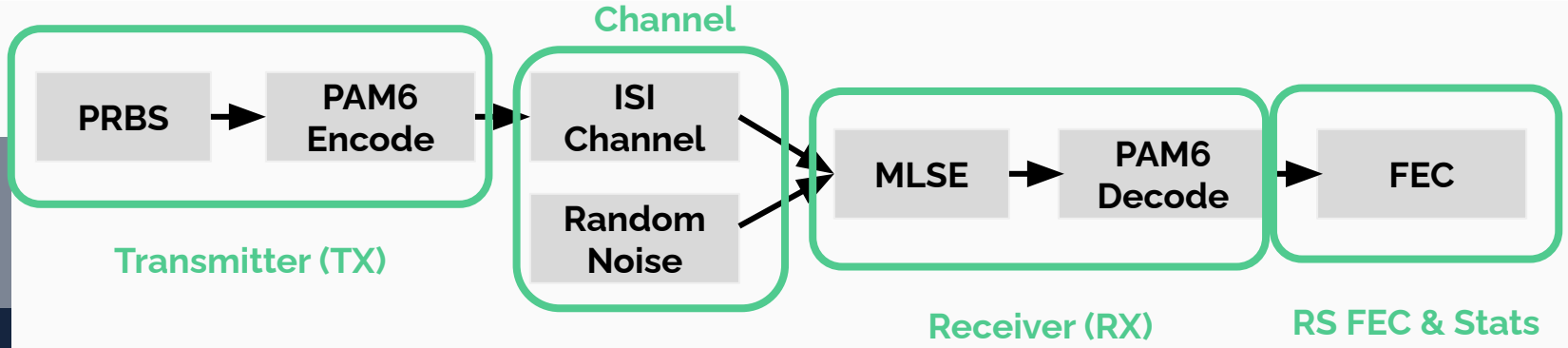
Supervisor: Prof. Tony Chan Carusone
Administrator: Mr. Matthew Yee

Agenda

1. Poster Presentation & Slides
2. Live Demo (AWS, Reference Model)
3. Q&A
4. Supervisor Evaluation
5. x
6. x

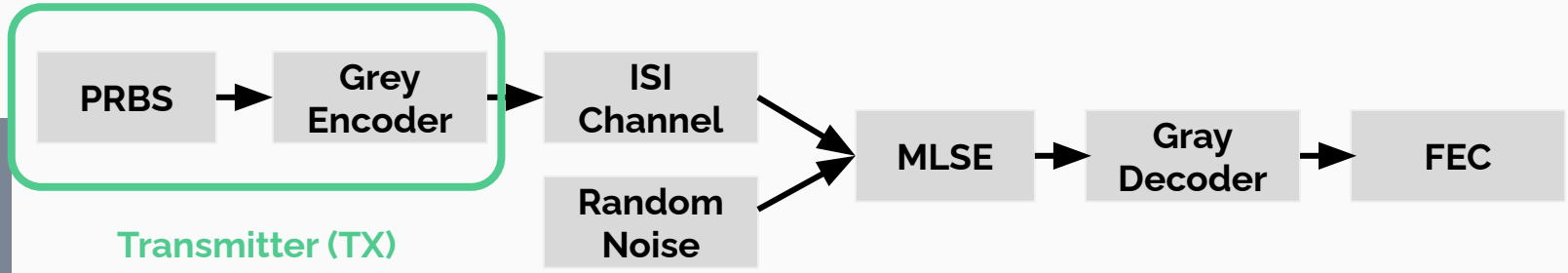
Variant 3: System 6





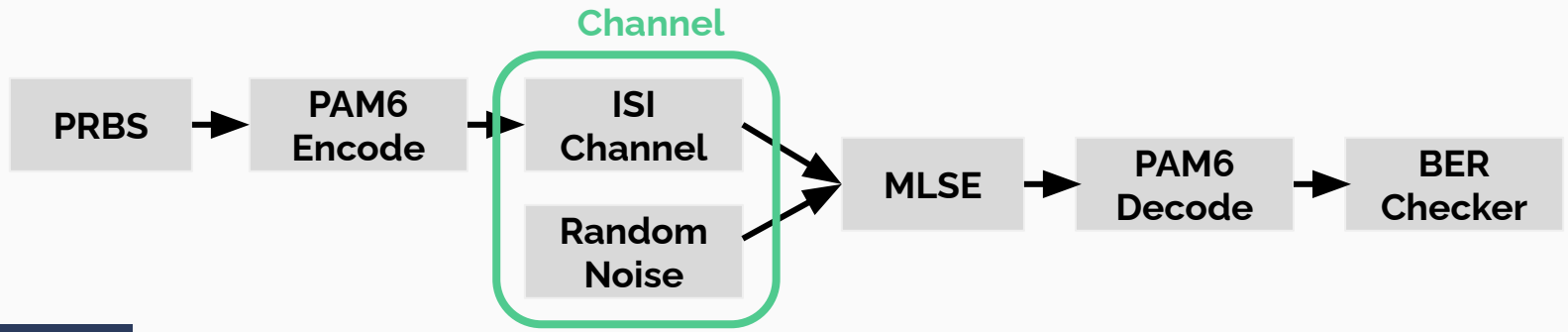
Reference Model

```
# System 6 Model: PRBS - preFEC
prbs = PRBS63(seed=args.seed)
py_bits = list(prbs.generate(args.n))
enc = GrayEncodePAM6()
syms = enc.encode(py_bits)
```



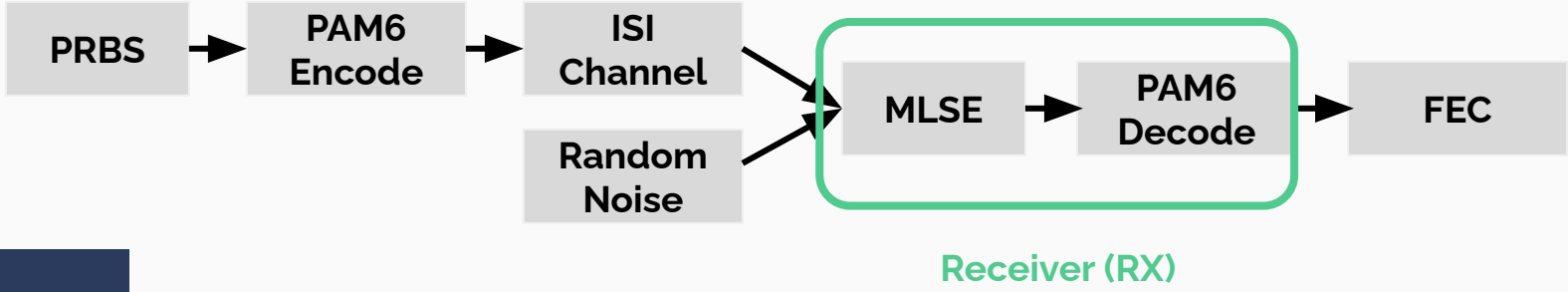
Reference Model

```
isi = ISIChannelOneTapPAM6(symbol_sep=24.0, alpha=0.5)
ch_out = isi.process(syms)

noise = Noise(mem_file="noise18dB_sep24.mem")
noise_vals, noisy, urng_rnd_vals, raw_sum_vals, sum_bits_vals, valids = noise.add_noise_burst_mode(ch_out)
```



Reference Model

```python
mlse = MLSEPAM6(SYMBOL_SEPARATION=24,SIGNAL_RESOLUTION=8,ALPHA=0.5,TRACEBACK=10,METRIC_RESOLUTION=20)
dec_syms = []
check_cnt = 0
for i in range(len(noisy)):
    if valids[i]:
        x8 = apply_8bit_wrap(int(noisy[i]))
        sym, valid = mlse.step(signal_in=x8, signal_in_valid=1, rstn=1)
        if valid:
            dec_syms.append(sym)
dec = GrayDecodePAM6()
py_post = dec.decode(dec_syms)
```

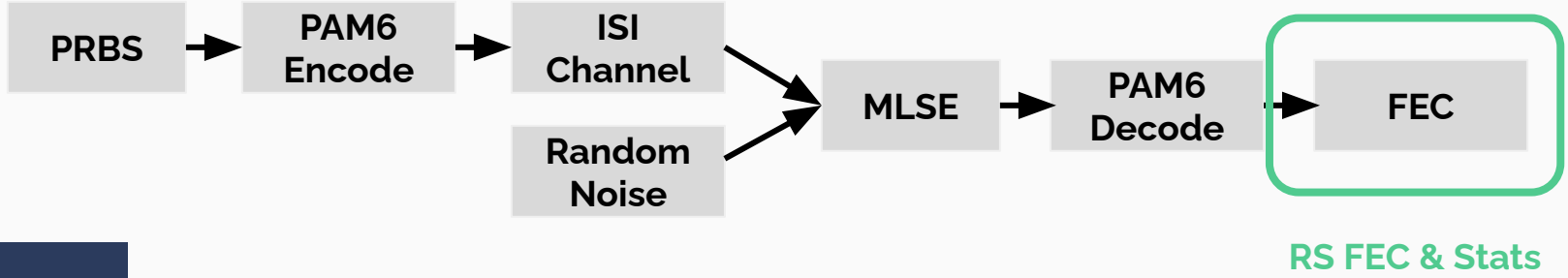


Zero Pre-FEC Mismatch for ~1M Bits

```
Celine@DESKTOP-B6PLPOU MINGW64 ~/Documents/4Winter/Capstone_Local/ece496-capstone/sysAll_PAM6/Python (ref_model)
$ python sys6_model_mismatch_demo.py
[RTL-PRBS vs Ref-PRBS] Compared 999933 bits (offset 0). Mismatches: 0  BER=0.000e+00
[RTL-PostGE vs Ref-PostGE] Compared 399971 bits (offset 0). Mismatches: 0  BER=0.000e+00
[RTL-PostISI vs Ref-PostISI] Compared 399970 bits (offset 0). Mismatches: 0  BER=0.000e+00
[RTL-PostMLSE vs Ref-PostMLSE] Compared 399959 bits (offset 0). Mismatches: 0  BER=0.000e+00
[RTL-PreFEC vs Ref-PreFEC] Compared 999890 bits (offset 0). Mismatches: 0  BER=0.000e+00
Reference simulation time: 6.14 seconds
```

Reference Model

```
fec = IL_FEC_Checker(seed=seed_val, n=544, t=15, m=10, n_interleave=1)
for bit in py_post:
    fec.step(bit)

ber_pre = fec.total_bit_errors_pre / max(1, fec.total_bits)
ber_post = fec.total_bit_errors_post / max(1, fec.total_bits)
```



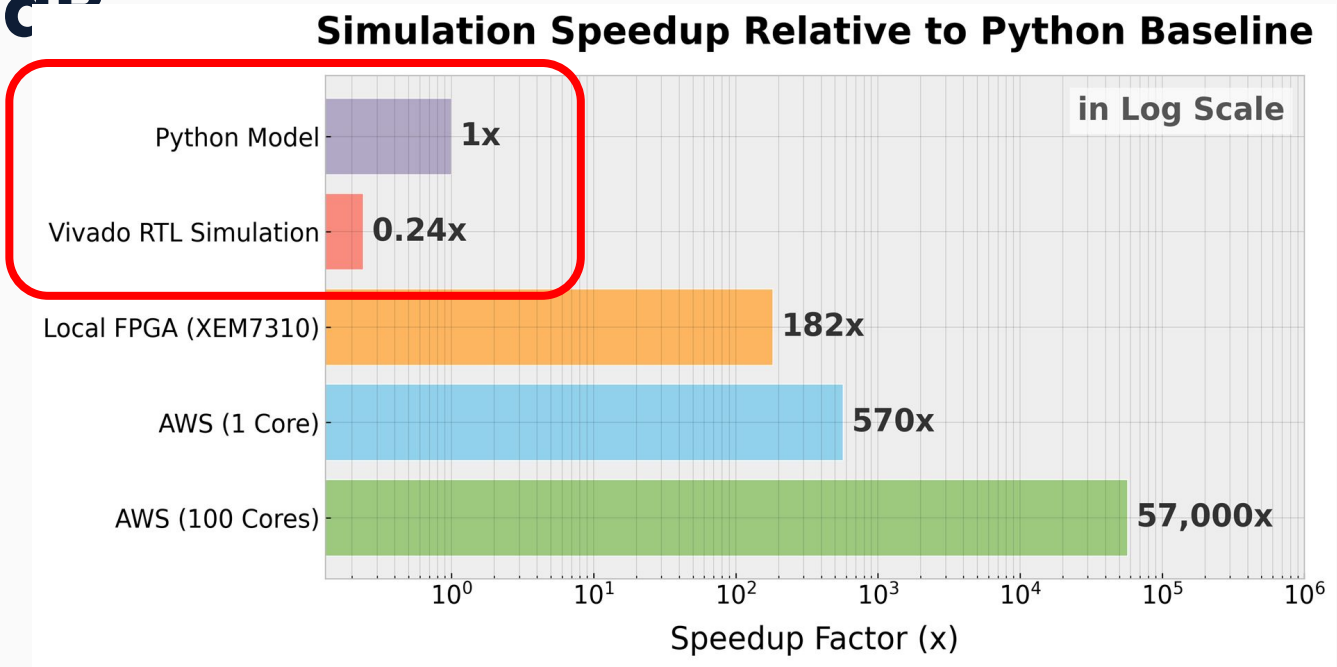
RTL vs Ref Model Post-FEC Matching

```
Celine@DESKTOP-B6PLPOU MINGW64 ~/Documents/4Winter/Capstone_Local/ece496-capstone/sysAll_PAM6/Python (ref_model)
$ python sys6_model_postFEC.py

RUNNING SIMULATION: 15 dB
Using noise memory: noise15dB_sep24.mem
[[15dB] RTL-PRBS vs Ref-PRBS] Compared 999933 bits (offset 0). Mismatches: 0  BER=0.000e+00
[[15dB] RTL-PreFEC vs Ref-PreFEC] Compared 999890 bits (offset 0). Mismatches: 0  BER=0.000e+00

Metric                    | RTL Value       | Python Ref      | Match?
-------------------------------------------------------------------------------
total_bits                | 999889          | 999970          | ✅
total_bit_errors_pre      | 46055           | 46061           | ✅
total_bit_errors_post     | 45879           | 45887           | ✅
total_frames              | 183             | 183             | ✅
total_frame_errors        | 183             | 183             | ✅
BER_pre                   | 4.606e-02       | 4.606e-02       | ✅
BER_post                  | 4.588e-02       | 4.589e-02       | ✅
Simulation time: 6.95 seconds
```

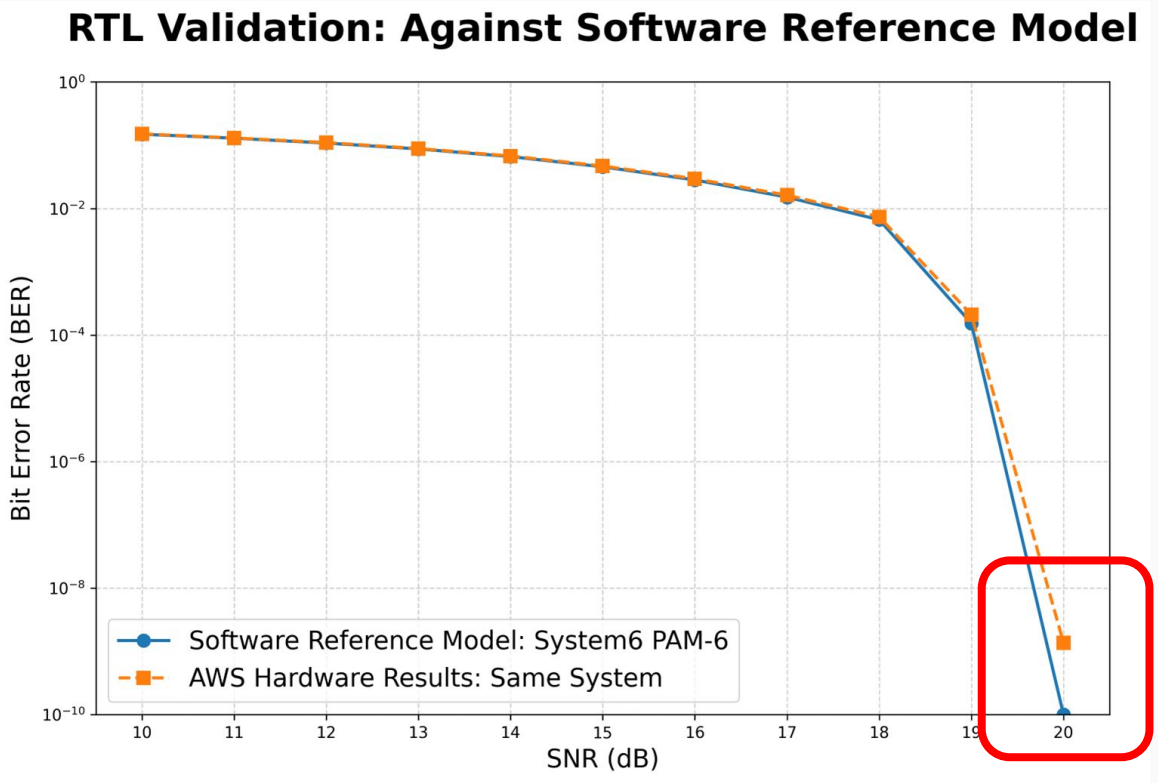
Slow simulation, 0 post-FEC errors @20dB

```
RUNNING SIMULATION: 20 dB
Using noise memory: noise20dB_sep24.mem
[[20dB] RTL-PRBS vs Ref-PRBS] Compared 999933 bits (offset 0). Mismatches: 0  BER=0.000e+00
[[20dB] RTL-PreFEC vs Ref-PreFEC] Compared 999890 bits (offset 0). Mismatches: 0  BER=0.000e+00

Metric                    | RTL Value       | Python Ref      | Match?
-------------------------------------------------------------------------------
total_bits                | 999889          | 999970          | ✅
total_bit_errors_pre      | 654             | 654             | ✅
total_bit_errors_post     | 0               | 0               | ✅
total_frames              | 183             | 183             | ✅
total_frame_errors        | 0               | 0               | ✅
BER_pre                   | 6.541e-04       | 6.540e-04       | ✅
BER_post                  | 0.000e+00       | 0.000e+00       | ✅
Simulation time: 6.84 seconds
```

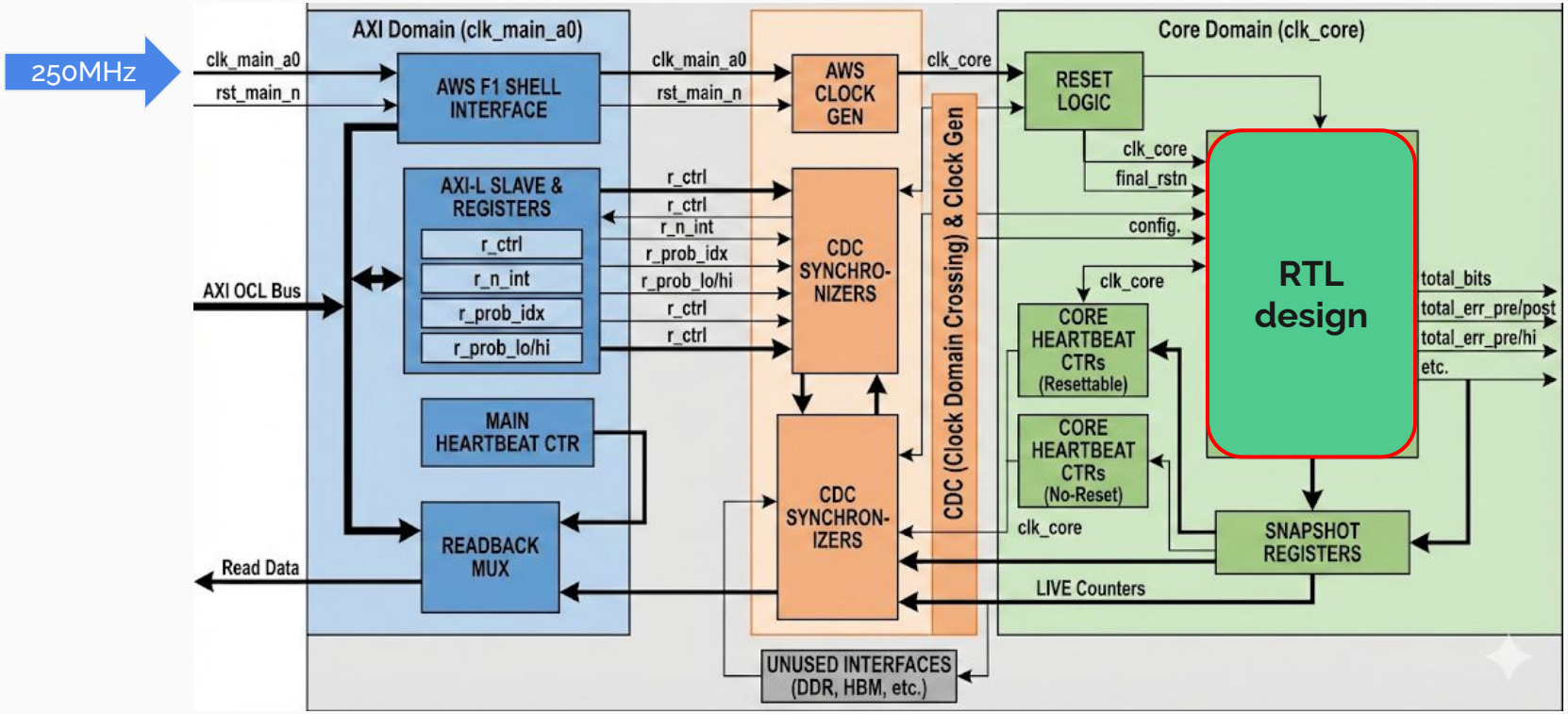
Slow simulation, 0 post-FEC errors @20dB



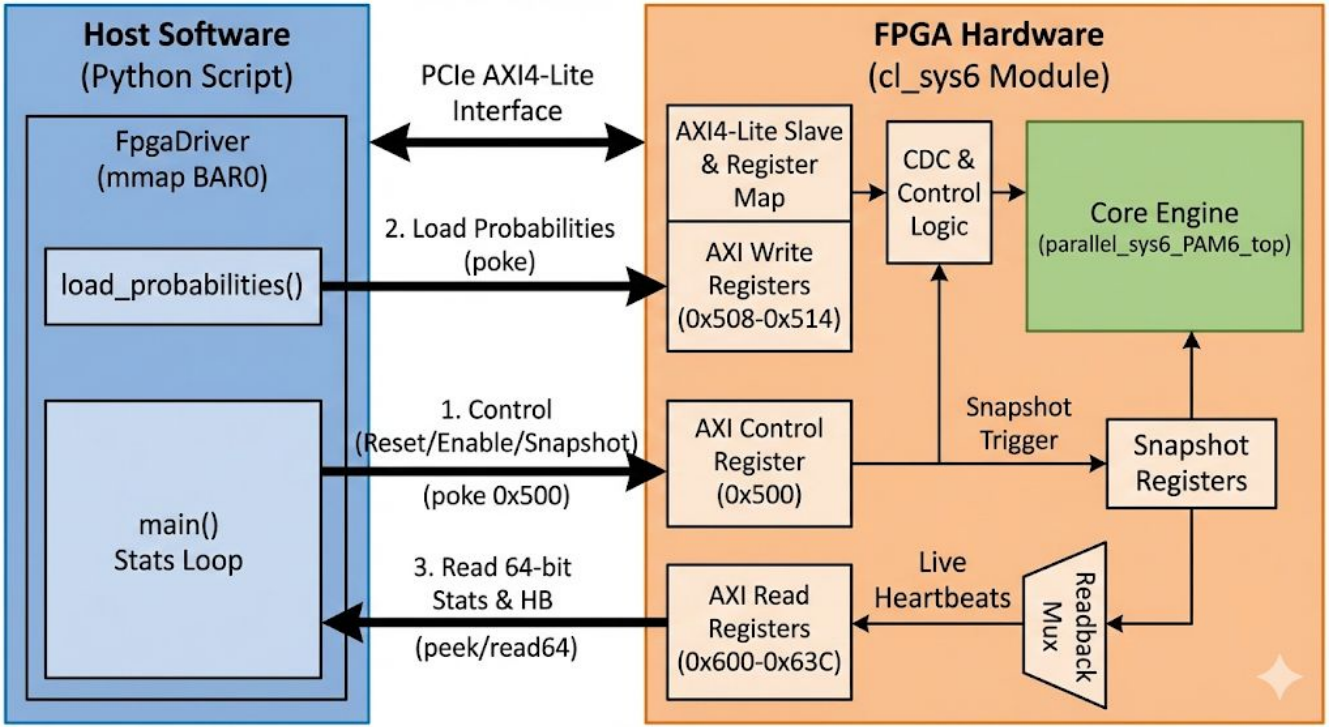
BER-SNR Comparison [Live Demo]



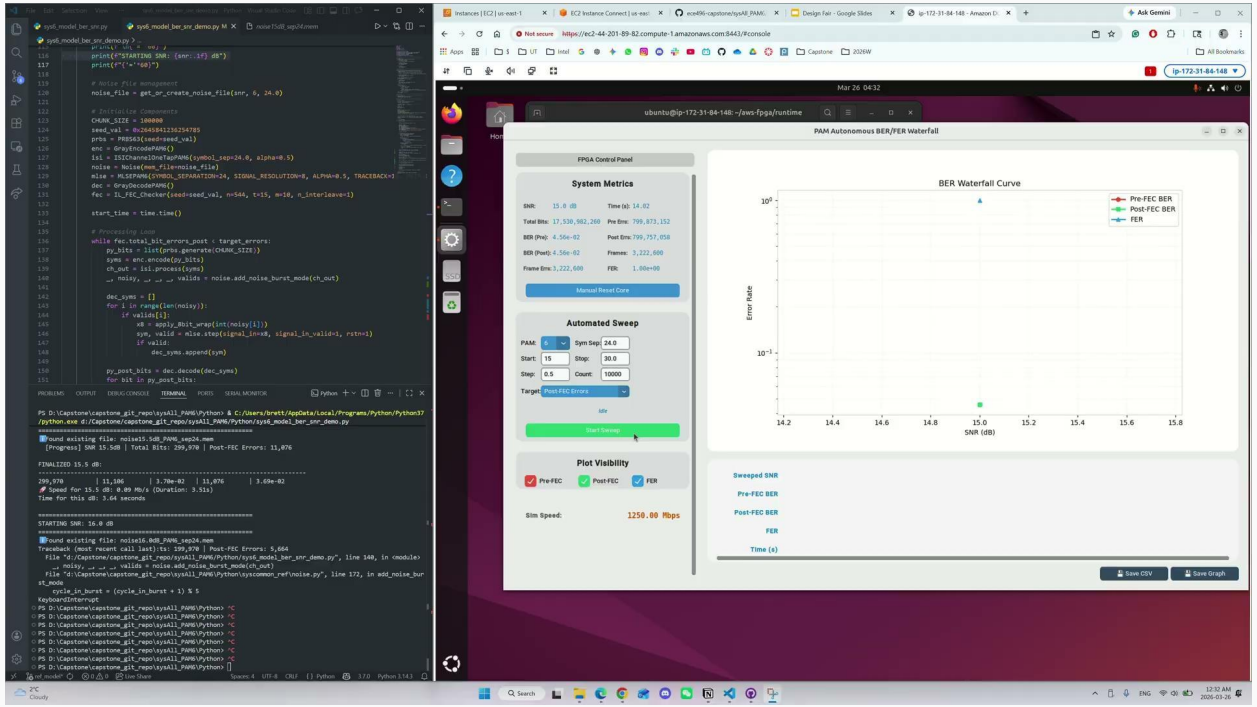
AWS Block Diagram: RTL Wrapper



AWS Block Diagram: Host Controller



Backup: Video Demo of Speedup



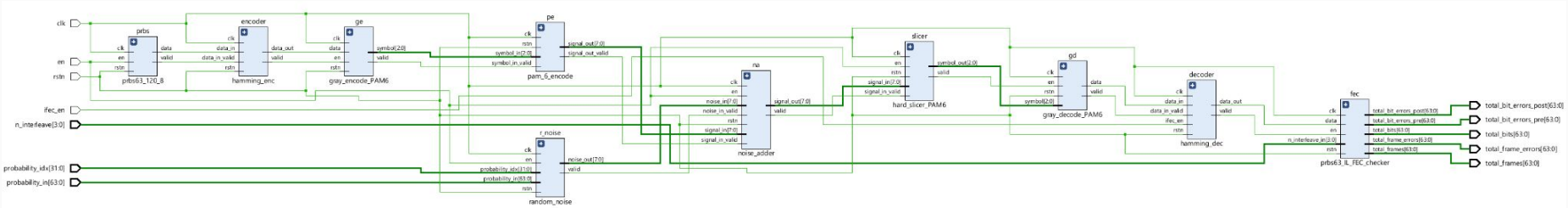
Cost Analysis

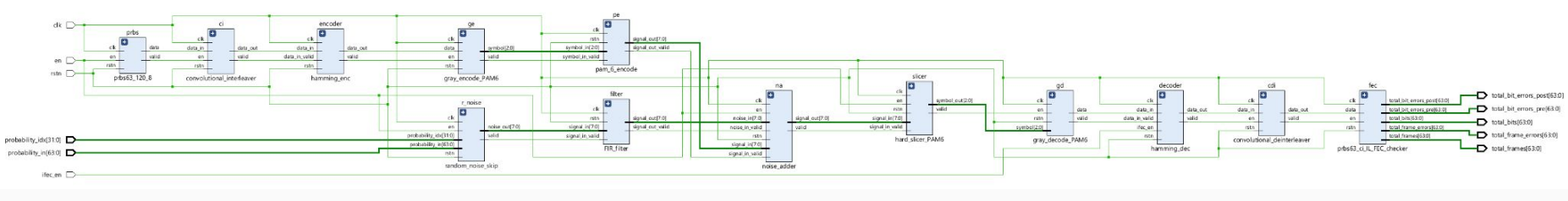
Platform	Speed (Mb/s)	Upfront Cost	Hourly Cost	Cost per 1T(10^{12}) bits
Software	0.219	$0	~$0.3 *	$380.52
Vivado RTL	0.052	$0	~$0.3 *	$1,602.56
Opal Kelly XEM7310	40	$630 [1]	$0	-
ZCU102 (Ref Design)	1200 [2]	$3,300 [3]	$0	-
AWS	125 * N_cores	$0	~$3 **	$6.67 / N_cores

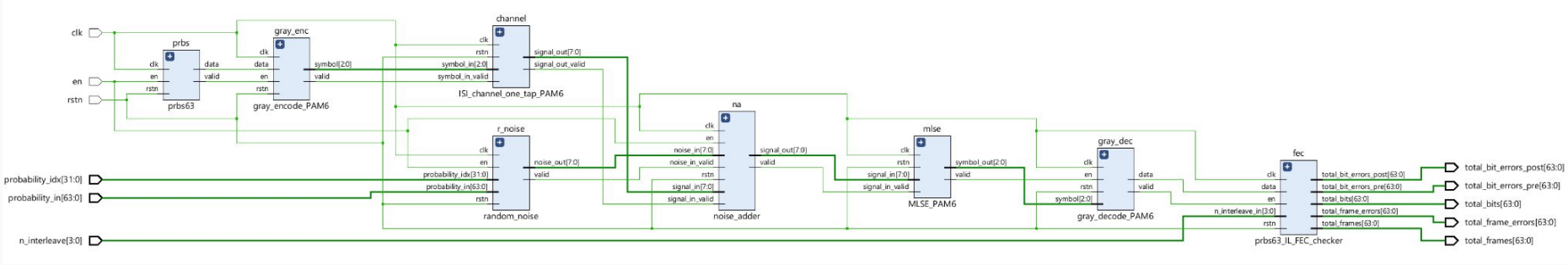
* AWS m7i-flex.xlarge (4 cores, 16GB memory, $0.20hr) + overhead (disk space, network fee)
** $1.98/hr for FPGA, ~0.50/hr for synthesis (assume 1hr run requires 1hr synthesis), + overhead

[1] https://www.opalkelly.com/products/fpga-integration/XEM7310
[2] https://utoronto.scholaris.ca/bitstreams/b231d761-ddc0-4073-8e21-c0baf1bd173a/download
[3] https://www.amd.com/en/products/adaptive-socs-and-fpgas/evaluation-boards/ek-u1-zcu102-g.html

Vivado RTL Block Diagrams

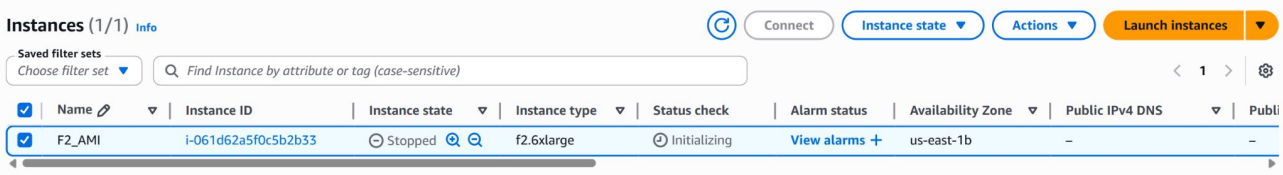






GUI Instructions: Setup

1. Launch F2 instance through AWS
   a. Click Instance state -> Start instance
   b. Once it is connected (3/3 status checks passed), check the instance & Click Connect



2. Follow steps 2 (Ubuntu GUI Setup) on User documentation to establish GUI on F2
3. Follow steps 4 (Run design on F2) on User documentation to get image that was synthesized and uploaded to cloud & run on F2
4. Run gui.py